FROM FOAM RUN TO CFD EVIDENCE

A practical post-processing guide for the cusped-leading-edge airfoil case

OpenFOAM Foundation 13 | shockFluid | Laminar, compressible, transient flow

THE CENTRAL IDEA

A solver run is not a result. Post-processing checks numerical behaviour, turns fields into physical quantities, extracts useful measurements, and then tests whether those measurements are trustworthy.

What this guide does

It recaps the case we have actually built, explains what the first run has and has not established, and lays out the next lessons in a practical order. The workflow is CLI-first: use OpenFOAM utilities and terminal-readable data first; use a visualiser only where a spatial contour genuinely helps.

Learning sequence

Stage	Question answered
1. Preserve and audit	Can I reproduce the run and inspect what happened?
2. Numerical health	Did the solver remain stable? Do residuals and Courant number make sense?
3. Derived fields	What are Mach number, speed and vorticity in the domain?
4. Flow structure	Where are shocks and strong gradients?
5. Profiles	What does pressure or Mach number do along a line?
6. Surface physics	How do we form Cp, drag, lift and moments?
7. Validation	Does the solution change with mesh, domain and simulation time?

Prepared from the observed case history and the OpenFOAM Foundation v13 User Guide. Commands are marked as first-run steps or later exercises where case-specific configuration still needs validation.

1. Where our case stands today

This is the actual state at the end of our setup session. Keeping these facts straight matters because the report must not claim more than the evidence supports.

Geometry and mesh history

Item	Observed state
Original STL	constant/triSurface/cusped_airfoil.stl; 2,388 triangles, 1,196 vertices; closed surface.
Transformed STL	cusped_airfoil_xz.stl was rotated into the X-Z profile plane. A further spanwise-scaled copy, cusped_airfoil_span1.stl, spans y = -0.5 to +0.5 m for the structured-mesh projection workflow.
Mesh method	Adapted the OpenFOAM Foundation 13 fluid/aerofoilNACA0012 tutorial: multi-block blockMesh, collapse span direction with transformPoints, then extrudeMesh to a thin 2D-style mesh.
Cell-count iterations	Initial structured mesh: 16,000 cells. Upstream cells increased: 20,800 cells. Further increase to xUCells=80: 24,000 cells.
Final checkMesh	24,000 hexahedra; 5 patches: aerofoil, inlet, outlet, back, front. Final summary said “Mesh OK”.
Mesh caveat	Maximum skewness 3.082 (passes), but maximum non-orthogonality 87.27 degrees and 186 faces above 70 degrees. This remains a serious quality warning for quantitative work.
Domain caveat	Generated mesh bounding box: x = -1.25 to 4 m and z = -2 to 2 m. This is much smaller and less symmetric than the paper’s reported 80-chord-diameter far field.

Solver setup and first run

- Solver module: foamRun with solver shockFluid, following the installed shockFluid/diffuserIntake tutorial as a starting point.
- Physics intended: laminar, compressible ideal gas, Sutherland viscosity; T_inf = 300 K; U_inf about 433.9 m/s along +X; p_inf = 1775.544 Pa.
- Boundary patches: aerofoil wall; inlet and outlet patches; front/back empty for a 2D-style model.
- Run controls: deltaT = 1e-7 s; endTime = 2e-5 s; writeInterval = 5e-6 s.

DO NOT OVERCLAIM

The run completed, but this is a startup demonstration—not yet a validated reproduction of the paper. The boundary conditions are a simplified adaptation of the tutorial, the domain is too small for direct paper comparison, and mesh non-orthogonality still needs work.

2. The post-processing workflow

OpenFOAM post-processing is not one command. It is a chain of checks, transformations and interpretation. The Foundation v13 guide groups this work into field calculation, graph sampling, force and coefficient calculation, probes, surfaces and other function objects [1-3].

Step	Action	Why it matters	Evidence to keep
0	Freeze a copy of the case and save the log.	Protects the current run from accidental edits and makes comparisons possible.	Case snapshot; log.foamRun.
1	Read solver health.	Detects blow-up, negative/unphysical fields, excessive Courant number or suspicious residual trends.	A short log summary.
2	Inspect saved times and fields.	Confirms which data actually exist and which time is being analysed.	List of time directories and field files.
3	Calculate derived fields.	Mach number, speed magnitude, vorticity and gradients reveal more than raw U, p and T alone.	Computed fields and command log.
4	Inspect spatial structure.	Locates shocks, expansion regions, shear layers and boundary-layer trends.	Contour or VTK output, plus interpretation.
5	Sample lines/surfaces.	Turns a coloured field into quantitative values that can be plotted and compared.	CSV or raw sample data.
6	Integrate wall forces.	Produces drag, lift and moment, then nondimensional coefficients.	Force/coefficient time histories.
7	Test validation.	Shows whether conclusions survive mesh, domain and time-step changes.	Comparison tables/plots.

A clean working directory check

```
cd ~/Projects/cusped-airfoil/openfoam-case
foamListTimes
ls -lh 2e-05/
ls -lh log.foamRun
```

The directories named 5e-06, 1e-05, 1.5e-05 and 2e-05 are saved time states. The directory 0 contains initial and boundary field files. The mesh lives in constant/polyMesh; physical properties and momentum transport live in constant; system contains run control, discretisation and linear-solver controls.

Keep post-processing separate from the original run log

```
foamPostProcess -help
foamPostProcess -list | grep -i -E 'MachNo|vorticity|graphUniform|forceCoeffsCompressible|
residuals|cellMax|cellMin'
```

Use the installed v13 list as the authority on what is available in this exact OpenFOAM distribution. The OpenFOAM.com/OpenCFD documentation and Foundation documentation are not interchangeable in every detail; this guide is written for Foundation v13 [1-3].

3. Lesson 1 — numerical health before physics

What the run log tells us

Log item	Plain-language meaning	Our first run
End	The solver reached the configured end time without reporting a fatal exit.	Reached 2e-05 s.
Courant Number max	Roughly, how far information travels across a cell per time step. It is a time-step stability/accuracy indicator, not an accuracy certificate.	Earlier tail: about 0.485.
Initial residual	How far the current equation solution is from its discretised linear-system solution at the start of that solve.	Ux around 4e-05; Uz around 3e-03; energy variable e around 3e-04 near the last steps.
Final residual	Residual remaining after the linear solver iteration in that step.	Tiny final residuals, often one linear iteration.
rho, no iterations	Density was handled via an algebraic/diagonal update in that solve. This is not automatically an error.	rho residual shown as zero, zero iterations.

Extract a compact run audit

```
grep -E 'Time =|Courant Number|Initial residual|Final residual|FOAM FATAL|nan|inf|^End' log.foamRun
| tail -n 100
```

How to interpret residuals

- Residuals measure the remaining imbalance in a discretised equation solve; they are not a direct measurement of physical error.
- Tiny final linear-solver residuals mean those linear systems were solved tightly. They do not prove that the whole flow is steady, grid-independent or physically correct.
- For transient flow, inspect trends across many time steps, field behaviour and forces. Do not require a time-dependent solution to become perfectly steady if the physics is inherently unsteady.
- Look for sudden spikes, divergence, `FOAM FATAL ERROR`, non-finite values, negative density/temperature/pressure, or fields that drift away from expected free-stream conditions.

Interpret our current result

The solver completed a 20-microsecond run and the maximum Courant number was about 0.485. That is enough to say the startup run progressed numerically; it is not enough to claim aerodynamic convergence.

4. Lesson 2 — check the physics and units

Before plotting the field, independently calculate what the undisturbed free stream should look like. This gives you a numerical expectation against which the simulation can be checked.

Freestream calculations for this case

Quantity	Relation	Approximate value
Specific gas constant	R = R_u / molecular weight	287.0 J/(kg K)
Sound speed	a_inf = sqrt(gamma R T_inf)	347.2 m/s
Speed for Mach 1.25	U_inf = M_inf a_inf	433.9 m/s
Density	rho_inf = p_inf/(R T_inf)	0.0206 kg/m^3
Dynamic pressure	q_inf = 0.5 rho_inf U_inf^2	about 1.94 kPa
Dynamic viscosity at 300 K	Sutherland law with As=1.458e-6, Ts=110.4 K	about 1.85e-5 Pa s
Reynolds number	Re = rho_inf U_inf c / mu_inf, c=1 m	about 4.85e5
Chord travel time	t_c = c / U_inf	about 2.30e-3 s

What this says about our run length

The current end time is t = 2.0e-5 s. Dividing by the chord travel time gives t/t_c about 0.0087, or 0.87% of one chord-travel time. The initial run is extremely short compared with a flow-through time across one chord.

Units to remember

- p: pascals (Pa) for this compressible case, not kinematic pressure p/rho.
- T: kelvin (K); U: metres per second (m/s); rho: kg/m^3; dynamic viscosity: Pa s.
- Mach number and pressure coefficient Cp are dimensionless.

Pressure is not Cp

The file `p` stores static pressure. A pressure-coefficient curve must be calculated from reference values: Cp = (p - p_inf) / q_inf, where q_inf = 0.5 rho_inf U_inf^2.

5. Lesson 3 — calculate Mach number

Mach number is local speed divided by local speed of sound. In a compressible flow it is more informative than velocity magnitude alone because the sound speed also changes with thermodynamic state:

Definition

M = |U| / a and for a calorically perfect gas, a = sqrt(gamma p/rho).

The Foundation v13 function object `MachNo` calculates and writes a volumetric Mach-number field [2,4]. First confirm availability in the installed release, then run it on the latest saved state.

```
foamPostProcess -list | grep -i MachNo
foamPostProcess -help
```

```
foamPostProcess -solver shockFluid -latestTime -func MachNo 2>&1 | tee log.postMach
```

The `-solver shockFluid` form is useful when the post-processing operation needs models/data constructed by the solver module. If the command reports an option or field error, do not edit the case at random: inspect `foamPostProcess -help` and the error message first [1].

Check what it wrote

```
find 2e-05 -maxdepth 1 -type f -printf '%f\n' | sort
find postProcessing -maxdepth 4 -type f | sort | tail -n 30
```

How to interpret Mach number

- Far upstream, expect the undisturbed value to be close to M = 1.25, subject to how the boundary conditions and derived field are set up.
- Across a shock, the flow normally shows a sharp change in Mach number; a shock also produces a pressure rise and thermodynamic changes.
- The field is not validated just because a contour looks plausible. Check the upstream numerical value and compare pressure/temperature changes too.
- A shock smeared over many cells can indicate insufficient resolution or dissipative numerical settings. A one-cell jump may also be a numerical artefact; check the scheme and mesh.

Other useful derived fields

Operation	What it tells you	Caution
mag(U)	Speed magnitude, independent of vector orientation.	Does not include sound speed.
vorticity	Curl of velocity; highlights rotational/shear regions.	Large values can be sensitive to mesh quality and gradients.
grad(p)	Pressure-gradient vector.	Strongly affected by shock resolution and mesh skewness.
totalEnthalpy	Energy-related scalar for compressible analysis.	Check thermo model assumptions before interpreting.

6. Lesson 4 — inspect the flow field without getting stuck in a GUI

A contour is useful when the question is spatial: where is the shock, where is pressure high, or where does the boundary layer separate? But a contour is not a substitute for data extraction and validation.

CLI-first path

```
foamPostProcess -solver shockFluid -latestTime -func MachNo
foamPostProcess -latestTime -func "mag(U)"
foamPostProcess -solver shockFluid -latestTime -func vorticity
```

Use the first command to create Mach number; the second computes velocity magnitude; the third requests vorticity. The exact output field names and successful model access should be checked in the command log and output directory. These are separate post-processing operations, so they should not overwrite the original fields.

When a contour is genuinely needed

```
foamToVTK -latestTime
```

This exports selected/latest case data into VTK files for use by a compatible viewer or a Python VTK/PyVista workflow. A visualiser is still needed to display spatial contours; if ParaView is frustrating, postpone that GUI step and use sampled line data first.

What to plot for this supersonic airfoil

- Mach number: locate the shock and assess how abrupt the change is.
- Static pressure p: identify pressure jumps across shocks and pressure distribution around the body.
- Temperature T: check compression-related heating and wall behaviour. The adiabatic wall condition should not force the wall temperature to remain at 300 K.
- Velocity magnitude: see acceleration/deceleration, but remember a vector plot is needed to understand direction.
- Vorticity or velocity gradients: inspect shear layers and near-wall changes, but do not overinterpret them on a mesh with maximum non-orthogonality near 87 degrees.

Don't make the first figure the conclusion

For every plot, write one sentence stating what feature you see, one sentence giving its physical interpretation, and one sentence stating what check is needed before trusting it.

7. Lesson 5 — turn the field into a line profile

A line sample converts a field into a table of values along a defined line. This is how a colour map becomes a plot you can measure, reproduce and compare. Foundation v13 includes graph functions such as `graphUniform`, which samples a specified number of points between a start and end location [2,3].

Choose the sampling line carefully

The airfoil profile lies in the X-Z plane; Y is the thin/extrusion direction. The current mesh is extruded from y = -0.1 to y = 0 m, so the mid-plane used in earlier diagnostics was y = -0.05 m. Do not blindly copy line coordinates from a tutorial that uses a different plane.

A centreline line at z = 0 passes through the solid airfoil between the leading and trailing edges. Therefore, split the centreline into an upstream segment and a downstream segment instead of sampling one line through the body.

Example `graphUniform` request: upstream centreline

```
# Example for later use as a function object; review the local template first
#includeFunc graphUniform(start=(-0.8 -0.05 0), end=(-0.02 -0.05 0), nPoints=200, fields=(p U))
```

The syntax above follows the Foundation v13 packaged function style. Add it through a `system/functions` file or execute through `foamPostProcess` after checking the local template. Start with the upstream segment, where the line is in fluid, and verify that the returned data contain the expected number of samples.

What to graph

Graph	Engineering question
p vs x upstream/downstream	Is the undisturbed pressure roughly correct? Is there a pressure disturbance upstream of the nose?
Mach vs x	Where does the solution change abruptly? Is the upstream value close to the target?
U magnitude vs x	Does the speed behave sensibly around the field of interest?
T vs x	Are temperature changes consistent with compression/expansion?

Why profiles are often better than a screenshot

A profile has numeric x and y values, exposes the thickness of a shock transition, can be plotted in Python, and can be compared across meshes or times. Record the line endpoints, sample count, field name, time and interpolation method with every profile.

8. Lesson 6 — pressure coefficient Cp

The paper discusses aerodynamic characteristics, so a useful later result is the surface pressure-coefficient distribution. Cp compares local pressure with free-stream dynamic pressure:

Pressure coefficient

Cp = (p_wall - p_inf) / (0.5 rho_inf U_inf^2)

With the current intended reference values, p_inf = 1775.544 Pa, rho_inf approximately 0.0206 kg/m^3 and U_inf = 433.9 m/s, the dynamic pressure is about 1.94 kPa. These are approximate consistency values, not measured output values.

What is needed to produce a trustworthy Cp plot?

1. Obtain pressure on the airfoil wall, not a line through the airfoil interior. The patch name is `aerofoil` in this case.
2. Check whether the surface-sampling function reports face-centre values or interpolated values. Record the method and location.
3. Use the same p_inf and q_inf reference consistently. Keep units in Pa until the nondimensionalisation step.
4. Plot Cp against x/c, with c = 1 m. For conventional airfoil plots, researchers often invert the Cp vertical axis; if you do, state it explicitly.
5. Check mesh convergence before interpreting small differences in Cp, particularly near the sharp leading edge.

Common mistakes

- Treating pressure in Pa as if it were Cp.
- Normalising with an arbitrary pressure or with the wrong density/velocity.
- Sampling cells in the fluid and calling it wall pressure without explaining the sampling method.
- Comparing a short startup snapshot against a time-averaged paper result.

Not yet the next command

We should not produce a final Cp plot until the pressure sampling method, mesh quality and reference values are made explicit. The formula is straightforward; trustworthy wall data are the difficult part.

9. Lesson 7 — forces, drag, lift and coefficients

The pressure field describes local loading. Integrating pressure and viscous stress over the airfoil wall gives forces and moments. For a compressible case, Foundation v13 provides `forcesCompressible` and `forceCoeffsCompressible` function objects [2,3].

Coefficient definitions

```
Cd = D / (0.5 rho_inf U_inf^2 A_ref)
Cl = L / (0.5 rho_inf U_inf^2 A_ref)
Cm = M_pitch / (0.5 rho_inf U_inf^2 A_ref c_ref)
```

The exact reference area and directions depend on the model orientation and the thickness used for the 2D-style extrusion. Here the flow is along +X, the profile thickness is along Z, and the extrusion is along Y. Drag direction is therefore +X; lift direction is usually Z; pitch axis is Y.

Configuration decisions before using forceCoeffsCompressible

- Patch: `aerofoil`, not inlet/outlet or the empty front/back patches.
- Reference velocity: approximately 433.9 m/s for the current alpha = 0-degree case.
- Reference chord: 1 m.
- Reference area: for the present finite extrusion depth of 0.1 m, a consistent 2D-style reference area would normally be chord × depth = 0.1 m^2. Confirm how the chosen force object treats the patch and depth before reporting coefficients.
- Reference pressure and density: define consistently with the compressible force function template and current absolute-pressure field.
- Time history: drag/lift coefficients need to be examined over time; one snapshot does not establish a stable average.

Hold off on final coefficients

Our current domain is too small for paper-level comparison and has severe non-orthogonality warnings. Learn the force-object setup, but do not submit the first coefficient as a validated value until the mesh, domain, BCs and reference area are checked.

10. Lesson 8 — time histories and monitoring

For a transient supersonic flow, the result is a sequence of fields in time. Post-processing should show how the solution evolves, not just what one time folder looks like.

Useful time histories

Signal	Purpose	Warning sign
Residuals per field	Shows linear equation solve behaviour over time.	Rapid growth or persistent increase; compare to field behaviour too.
Maximum Courant number	Tracks how aggressive the time step is relative to the cell scale and velocity.	Unexpected jumps or values beyond the planned limit.
p, T, U at probes	Tracks a local shock passage or transient wave.	Non-physical spikes, drift, or a signal that changes drastically with mesh/time step.
Cd, Cl, Cm	Tracks integrated aerodynamic loading.	Continual drift means no stable mean yet; oscillations may be physical or numerical.
Mass flow at boundaries	Checks consistency of mass flux in/out.	Large unexplained imbalance or a boundary condition that is not appropriate.

Useful OpenFOAM functions to discover

```
foamPostProcess -list | grep -i -E 'residuals|probes|graphUniform|patchFlowRate|
forceCoeffsCompressible'
```

Function objects can run after the solver or during it. Post-run processing is best for exploring existing data; run-time functions are best when you already know which measurements you want and wish to collect them automatically during a future long run [1,2].

How long should the simulation run?

There is no universal physical time. Estimate a convective time, c/U_inf, then run long enough to see the transients of interest and inspect whether measured quantities settle or develop repeatable periodic behaviour. For this case c/U_inf is around 2.30 ms, while the first run reaches only 0.020 ms.

That ratio is a scale estimate, not a magic criterion. Shock settling, domain waves, boundary reflections, and viscous development can require multiple characteristic times. Check the literature and the physical behaviour before setting the final end time.

11. Lesson 9 — validation before comparison with the paper

A plausible-looking contour is not validation. The goal is to show that the conclusions are not artefacts of mesh, domain, time step or boundary treatment.

Current risks to fix or quantify

Issue	Current evidence	Why it matters / next action
Mesh non-orthogonality	Max 87.27 degrees; 186 faces > 70 degrees.	Can degrade gradient/flux accuracy and make shock predictions sensitive to numerics. Improve leading-edge/block geometry or demonstrate robustness with refinement and sensitivity tests.
Far-field extent	Bounding box x = -1.25 to 4 m, z = -2 to 2 m.	The paper uses a much larger circular far field (diameter 80 chord lengths in final simulations). Boundary influence may contaminate waves/shocks.
Structured grid resolution	24,000 cells; 220 faces on aerofoil patch.	Not the paper's approximately 425×315 O-grid. Near-wall spacing and resolution need checking.
Boundary conditions	Simple fixedValue inlet and zeroGradient outlet; wall noSlip and zeroGradient T.	These are not identical to Fluent pressure-far-field conditions. Verify that boundaries suit the supersonic flow and layout.
Run duration	End time 2e-5 s; t/t_c about 0.0087.	Far too short to claim established transient or mean aerodynamic statistics.
Physical property simplifications	Sutherland viscosity; Cp/Pr choices may not match all paper settings exactly.	Record and justify property assumptions; conductivity/Prandtl details may need additional modelling.

A minimum validation plan

6. Mesh sensitivity: compare at least two or three grids, refining the airfoil and especially the leading edge while tracking Cp and force coefficients.
7. Domain sensitivity: enlarge the far field and compare shock/pressure pattern and wall loads.
8. Time-step sensitivity: reduce deltaT and check whether sampled fields and force histories change materially.
9. Boundary-condition sensitivity: verify correct freestream state and avoid artificial reflections as far as possible.
10. Physics comparison: verify expected upstream Mach number, shock pressure rise, temperature trends and force sign/magnitude.

Meaning of Mesh OK

The current `checkMesh` ended with "Mesh OK" because the skewness check passed. It still reported maximum non-orthogonality of 87.27 degrees and 186 faces above 70 degrees. Preserve that warning in the record; "Mesh OK" is not a certificate of numerical accuracy.

12. Lesson 10 — turn analysis into an engineering report

For every figure/table, record five things

11. Question: what are we trying to learn? Example: “Is the undisturbed flow close to Mach 1.25?”
12. Method: which field/function, time, patch or sample line produced the data?
13. Result: report units, extrema or characteristic values, and the figure/table.
14. Interpretation: explain the physics in plain language, separating observation from explanation.
15. Limitation: state which mesh/domain/time-step or BC uncertainty may affect the result.

Reproducibility habits

```
mkdir -p notes logs
cp log.foamRun logs/log.foamRun.first-startup
foamListTimes > notes/time-directories.txt
checkMesh > logs/checkMesh.final.txt 2>&1
```

Before a substantial change, preserve the current setup files. If the project is under Git, use a clean commit for a meaningful milestone. Never overwrite the only copy of a successful run or change several unrelated settings at once; otherwise it becomes hard to identify which change caused an effect.

A professor-ready summary of the present milestone

“I began from the OpenFOAM Foundation 13 `fluid/aerofoilNACA0012` structured-mesh tutorial and the `shockFluid/diffuserIntake` solver tutorial. I adapted the airfoil geometry and mesh blocks to the cusped profile, performed iterative mesh refinement, and configured a transient laminar compressible ideal-gas case. The first `shockFluid` run completed to 2×10^-5 s and wrote multiple time states. This demonstrates the case can initialise and advance. Post-processing and validation are now being used to check Mach-number recovery, shock structure, wall pressure/forces, and sensitivity to mesh, domain and time step. The current grid still has a high maximum non-orthogonality and the domain is smaller than the paper’s; results are therefore preliminary.”

The lesson checklist

Lesson	You should be able to answer
1. Numerical health	Did the run complete and behave stably?
2. Physical sanity	Does the upstream flow match M, p, T, rho, Re and units?
3. Mach number	Where is the flow subsonic/supersonic, and where do shocks appear?
4. Spatial fields	What structure is visible in p, T, |U| and vorticity?
5. Line sampling	Can I produce a repeatable line profile with coordinates and units?
6. Cp	Can I derive a meaningful dimensionless surface pressure curve?
7. Forces	Can I report Cd/Cl/Cm with correct axes and reference area?
8. Time histories	Are probes/loads settling or showing repeatable unsteadiness?
9. Validation	Are results robust to mesh, domain and time-step changes?
10. Reporting	Can another person repeat the workflow and see the evidence?

13. The next commands for our case

Do these in order, and pause to inspect the output after each stage. The guide intentionally does not automate force coefficients yet; the mesh/domain caveats should be addressed first.

A. Record the present state

```
cd ~/Projects/cusped-airfoil/openfoam-case
foamListTimes
ls -lh 2e-05/
grep -E 'Time =|Courant Number|FOAM FATAL|^End' log.foamRun | tail -n 50
```

B. Discover and compute Mach number

```
foamPostProcess -list | grep -i MachNo
foamPostProcess -help
foamPostProcess -solver shockFluid -latestTime -func MachNo 2>&1 | tee log.postMach
find 2e-05 -maxdepth 1 -type f -printf '%f\n' | sort
```

If the utility complains, stop and keep the exact error. Do not create new fields by hand to bypass it. The next step is to resolve the exact v13 function-object/solver context based on the message.

C. Optionally calculate speed magnitude and vorticity

```
foamPostProcess -latestTime -func "mag(U)" 2>&1 | tee log.postMagU
foamPostProcess -solver shockFluid -latestTime -func vorticity 2>&1 | tee log.postVorticity
```

D. Keep the results traceable

```
find 2e-05 -maxdepth 1 -type f -printf '%f %s bytes\n' | sort
find postProcessing -maxdepth 5 -type f -printf '%p\n' | sort | tail -n 80
```

First milestone after this guide

Calculate Mach number, verify the freestream is close to 1.25, and extract one clean pressure/Mach line profile outside the solid airfoil. Only after those checks should we configure and interpret Cp and integrated forces.

References

[1] CFD Direct / OpenFOAM Foundation, OpenFOAM v13 User Guide, Chapter 7: Post-processing. https://doc.cfd.direct/openfoam/user-guide-v13/postprocessing

[2] CFD Direct / OpenFOAM Foundation, OpenFOAM v13 User Guide, 7.2: Post-processing command-line interface. https://doc.cfd.direct/openfoam/user-guide-v13/post-processing-cli

[3] CFD Direct / OpenFOAM Foundation, OpenFOAM v13 User Guide, 7.3: Post-processing functionality (MachNo, graphUniform, forces, probes and surface functions). https://doc.cfd.direct/openfoam/user-guide-v13/post-processing-functionality

[4] OpenFOAM Foundation, v13 source guide: MachNo function object. https://cpp.openfoam.org/v13/classFoam_1_1functionObjects_1_1MachNo.html

[5] OpenFOAM Foundation v13 tutorial used locally as mesh reference: $FOAM_TUTORIALS/fluid/aerofoilNACA0012.

[6] OpenFOAM Foundation v13 tutorial used locally as solver reference: $FOAM_TUTORIALS/shockFluid/diffuserIntake.

[7] Akram, S. and Hasan, N. (2024), “The dynamics of supersonic flow past a new cusped leading edge airfoil”, Progress in Computational Fluid Dynamics, 24(1), 1–28. DOI: 10.1504/PCFD.2022.10052925.